



ME 310 Project: Thermo Fluid System Design

DESIGN AND FABRICATION OF AN STHX WITH DISK AND DOUGHNUT SHAPED BAFFLES

Group -05 (B2)



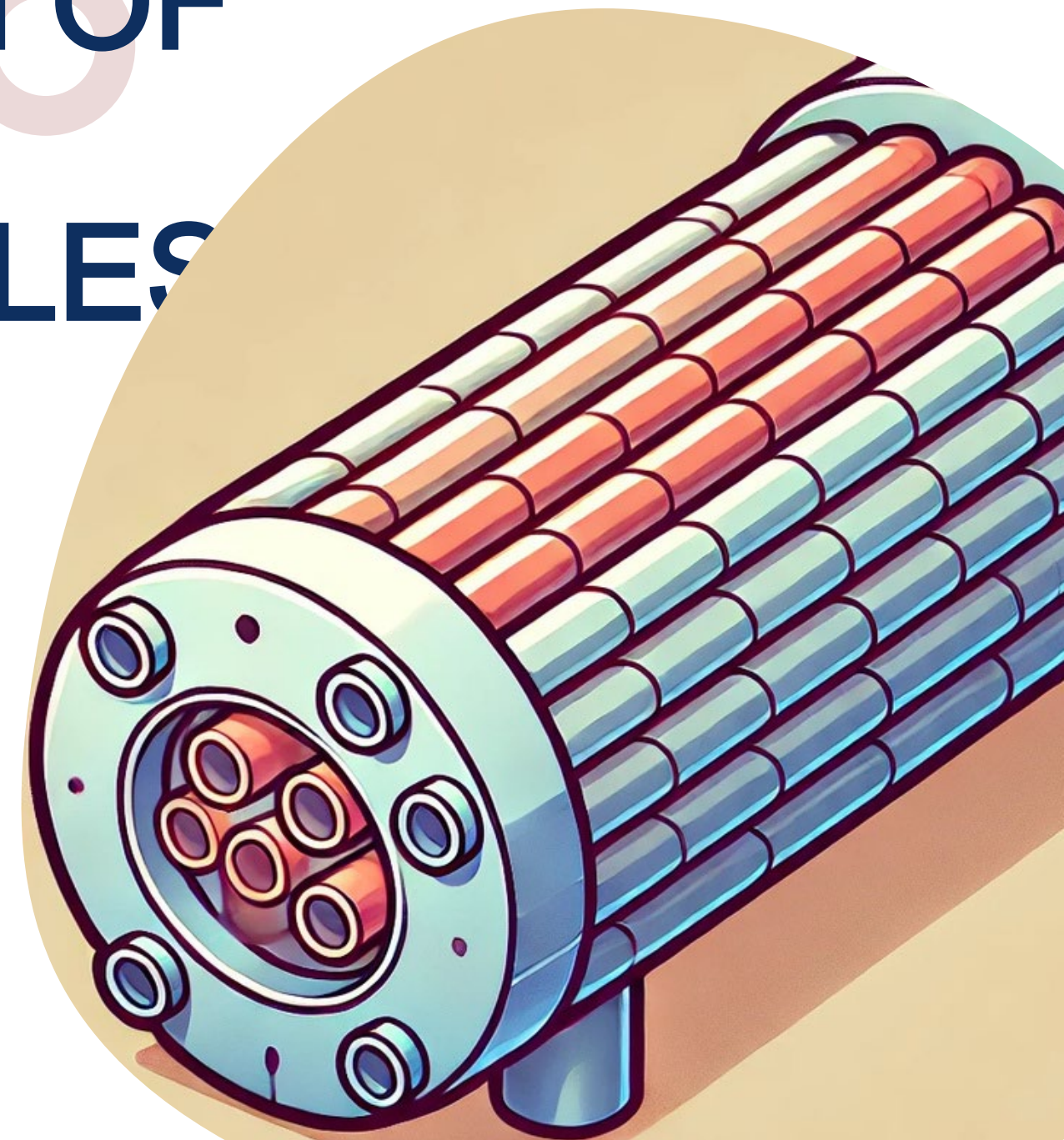
Team Members:

2010091: Md. Khalilur Rahman

2010093: Ibnul Muhtadi Shah

2010104: Md. Amanullah Asif

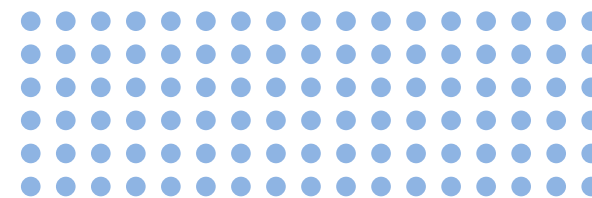
2010119: Kazi Md. Zareer



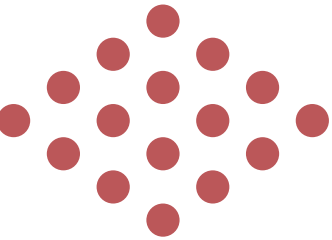


Introduction

- **Wastewater Heating Problem** : Food processing plants generate large volumes of heated wastewater ($\sim 70^{\circ}\text{C}$) during cleaning, sterilization, and pasteurization.
- **Environmental Concerns** : Releasing hot wastewater harms aquatic ecosystems and violates temperature limits set by environmental regulations.
- **Need for Cooling** : Cooling wastewater to $55\text{--}60^{\circ}\text{C}$ ensures safe discharge and environmental compliance.



Problem Statement



Problem

Heated wastewater (70°C) from food processing plants must be cooled before discharge to meet environmental regulations .

Objective

Design a Shell-and-Tube Heat Exchanger (STHX) to cool wastewater from 70°C to 60°C using cooling water at 25°C , both flowing at 0.3 kg/s .

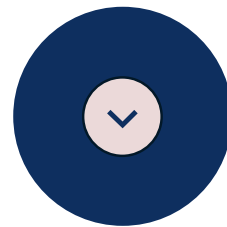


Constraints

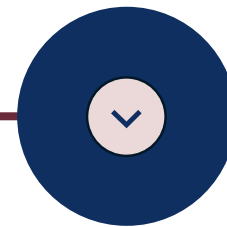
- Maximize heat transfer efficiency .
- Minimize pressure drop on the shell side .

Project Overview

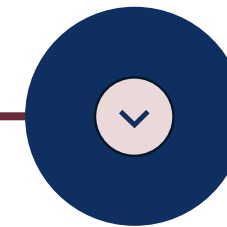
STHX is widely used for its robust design, ability to handle high pressures, and efficient heat transfer. It is chosen because of its high efficiency, durability, flexibility, and ease of maintenance. **Modifying the baffle configuration** improves the heat exchanger's performance by:



Enhancing turbulence to increase heat transfer.

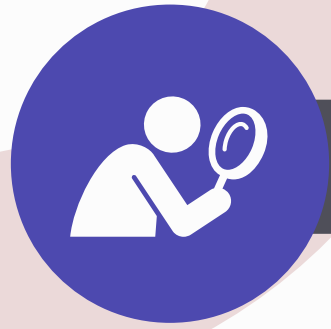


Reducing dead zones to ensure more uniform flow



Optimizing flow paths to minimize pressure drop

DESIGN ROADMAP



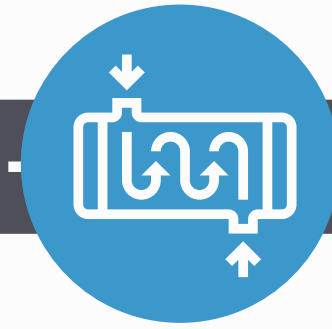
Problem Identification

Defined the performance issues in conventional STHX designs



Literature Review

Conducted review on existing research



Project Selection

Selected STHX with Disk-and-doughnut baffles



Analytical Calculations

Performed hand calculations

3D Modeling & CFD Simulation

Used Solidworks Modeling and Flow Simulation



Manufacturing

Fabricated the STHX using locally available materials after market analysis



Performance Testing

Performed Pressure and Leakage Tests

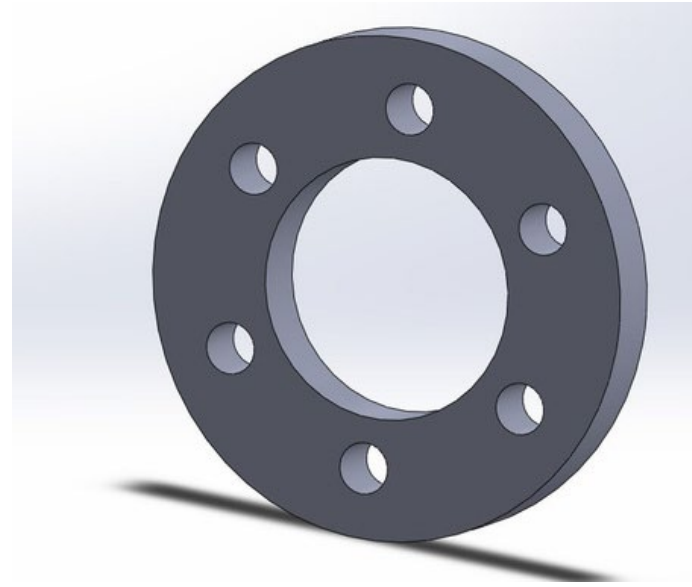
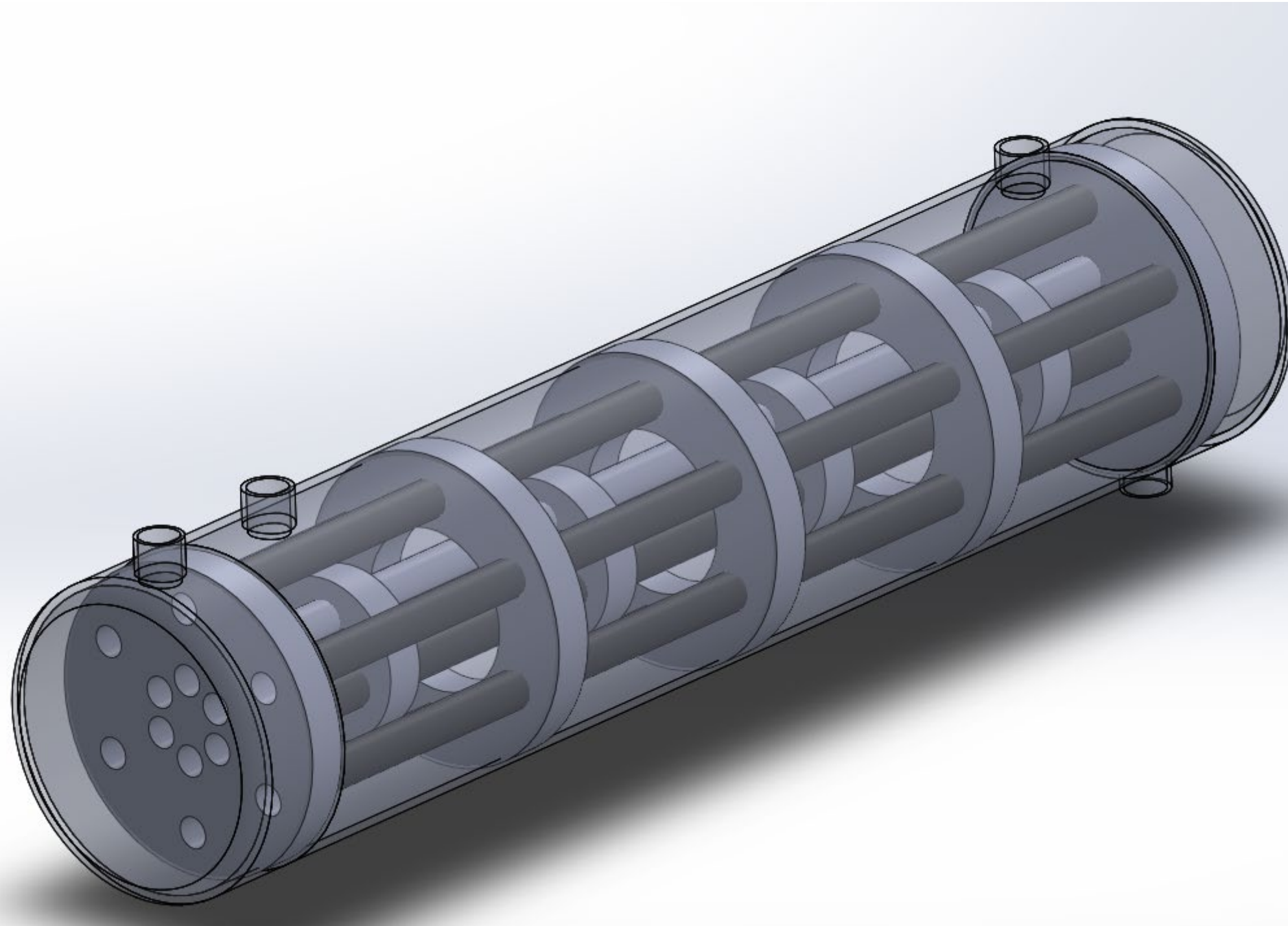


Report Writing

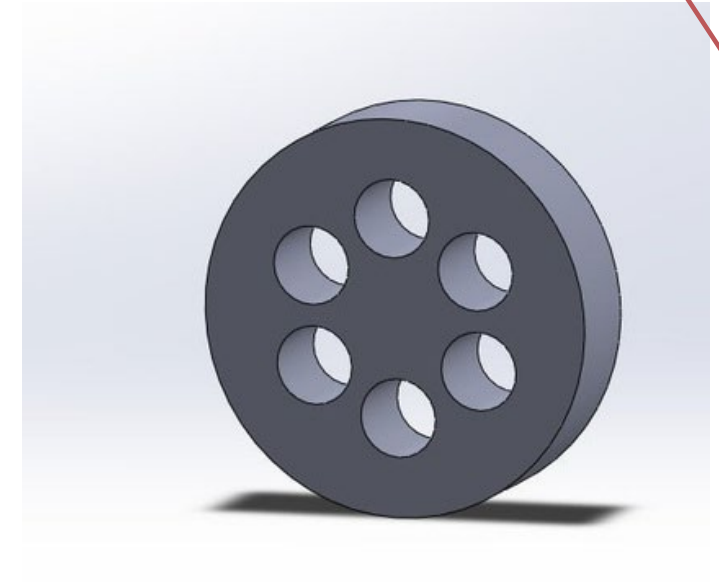
Submitted the report which included detailed calculations, procedures, and suggestions for improvement



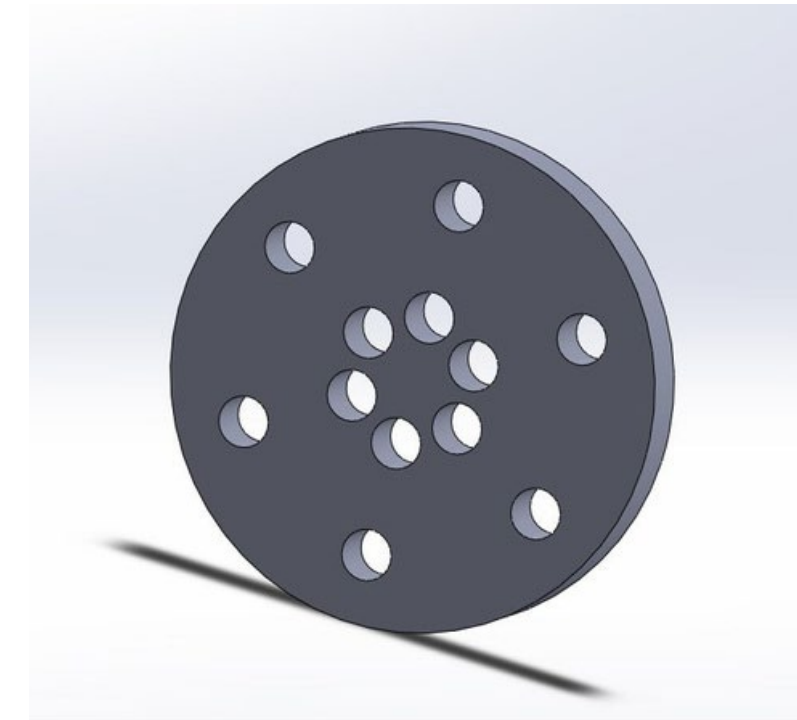
SolidWorks Model



Disk

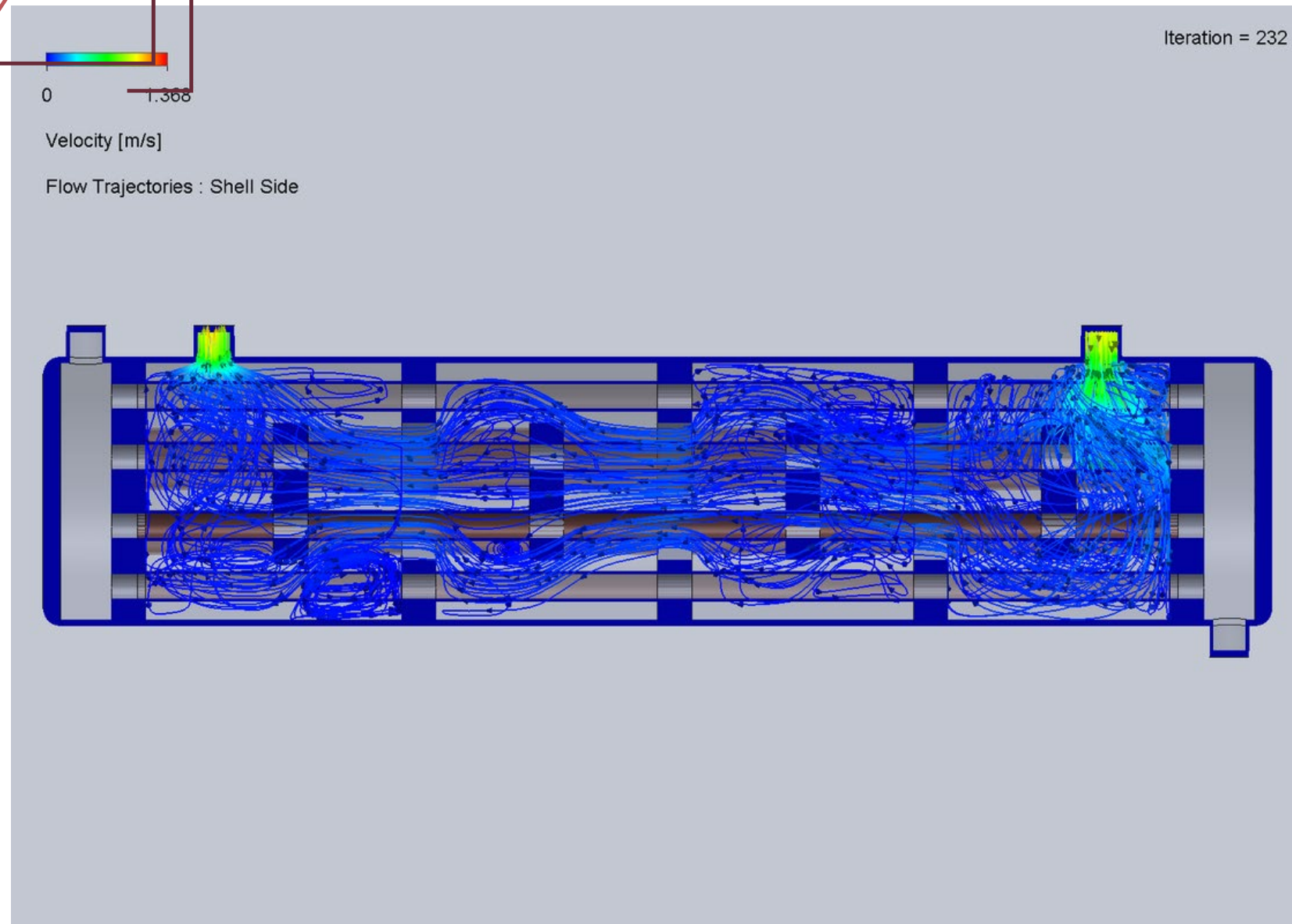


Doughnut

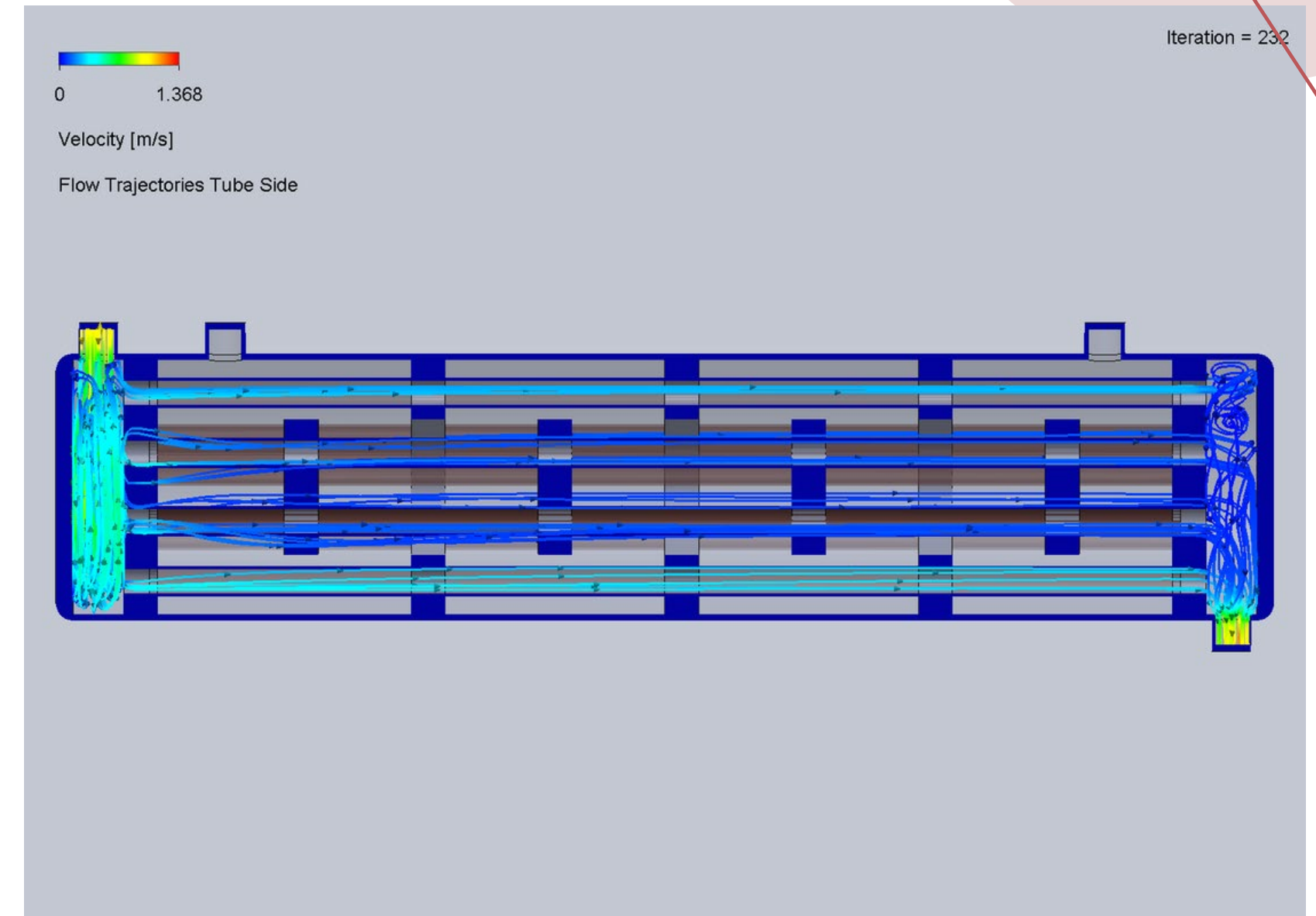


Gasket

Flow Simulation using SolidWorks

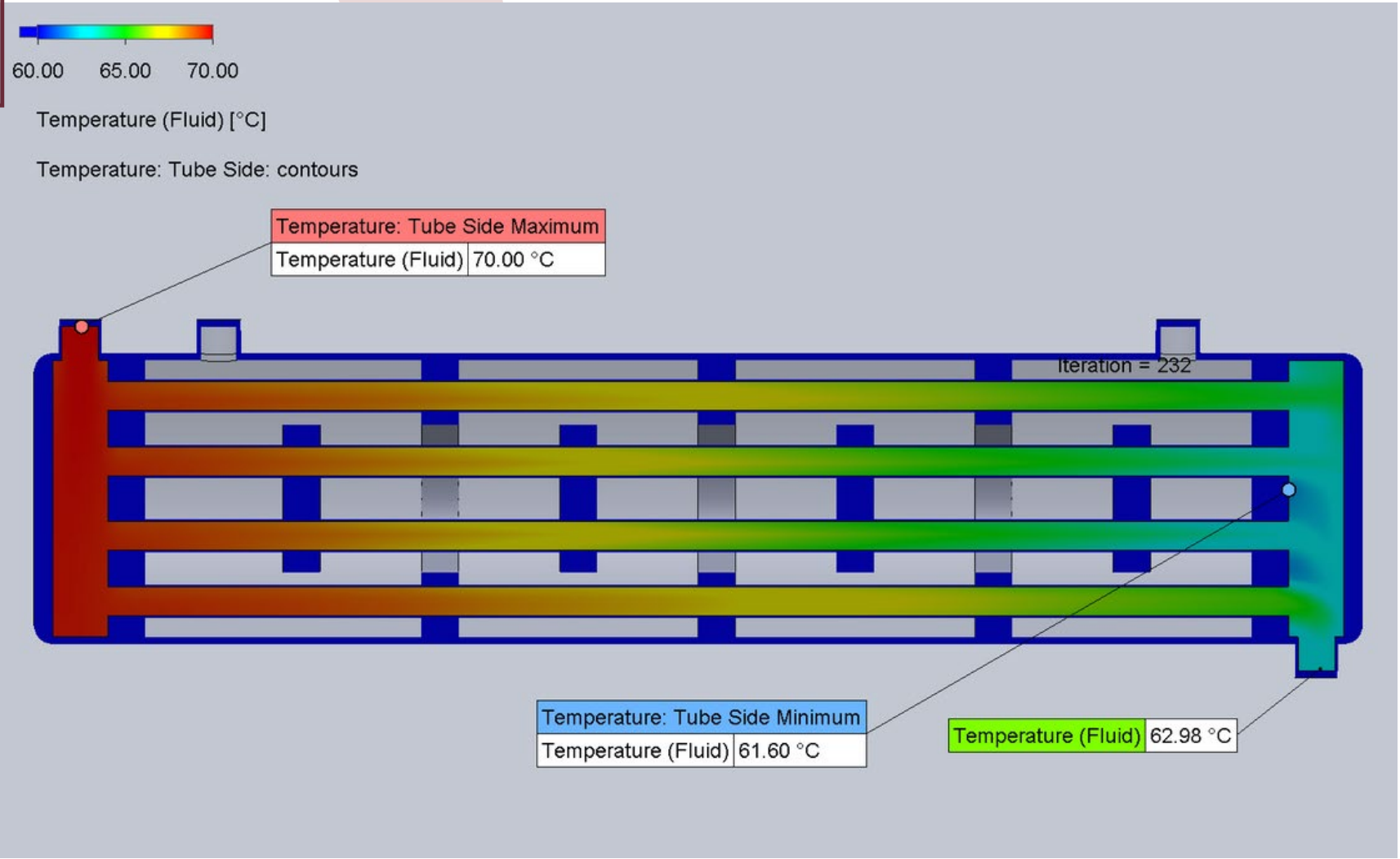


Flow Trajectory: Shell Side



Flow Trajectory: Tube Side

Flow Simulation using SolidWorks



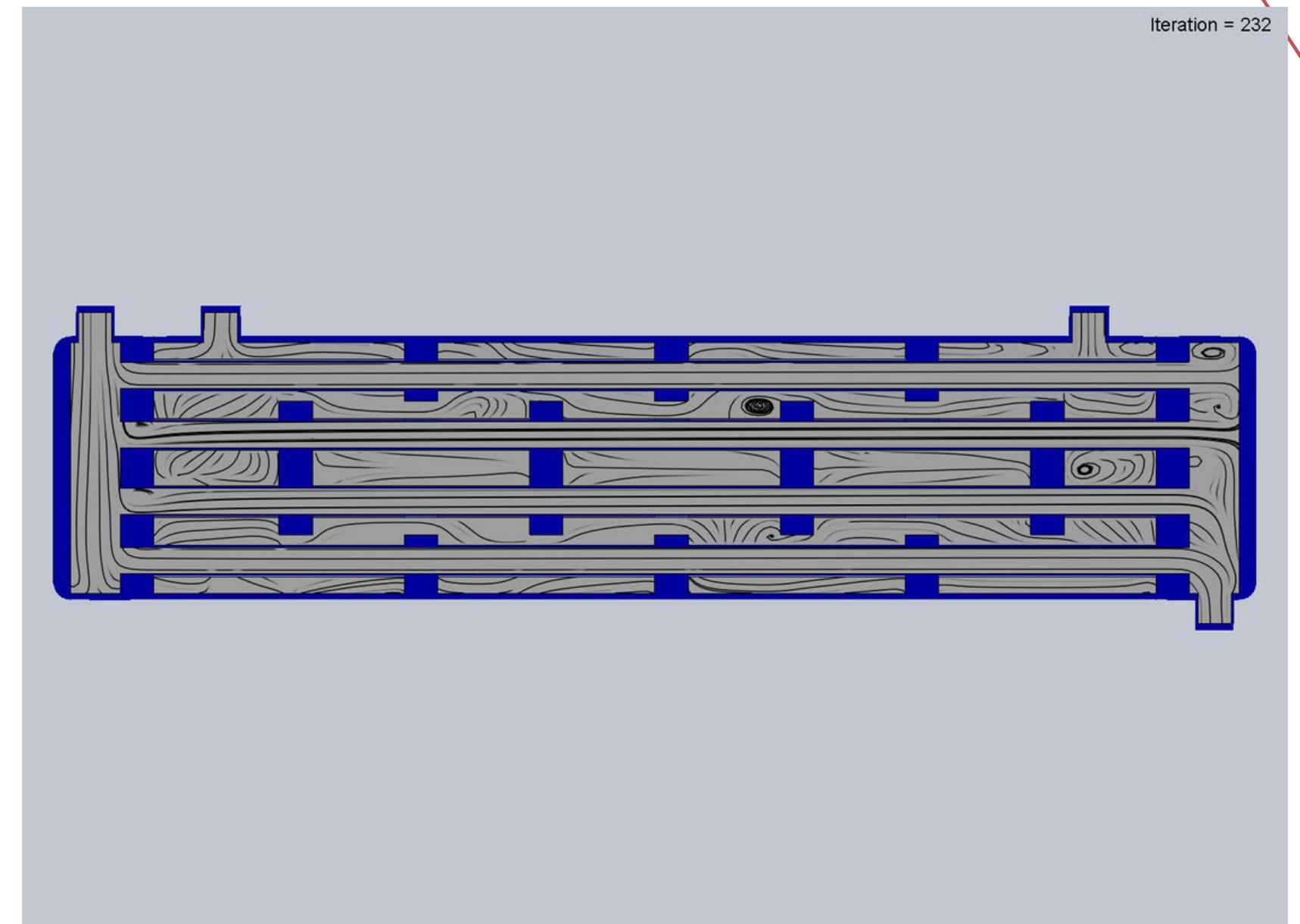
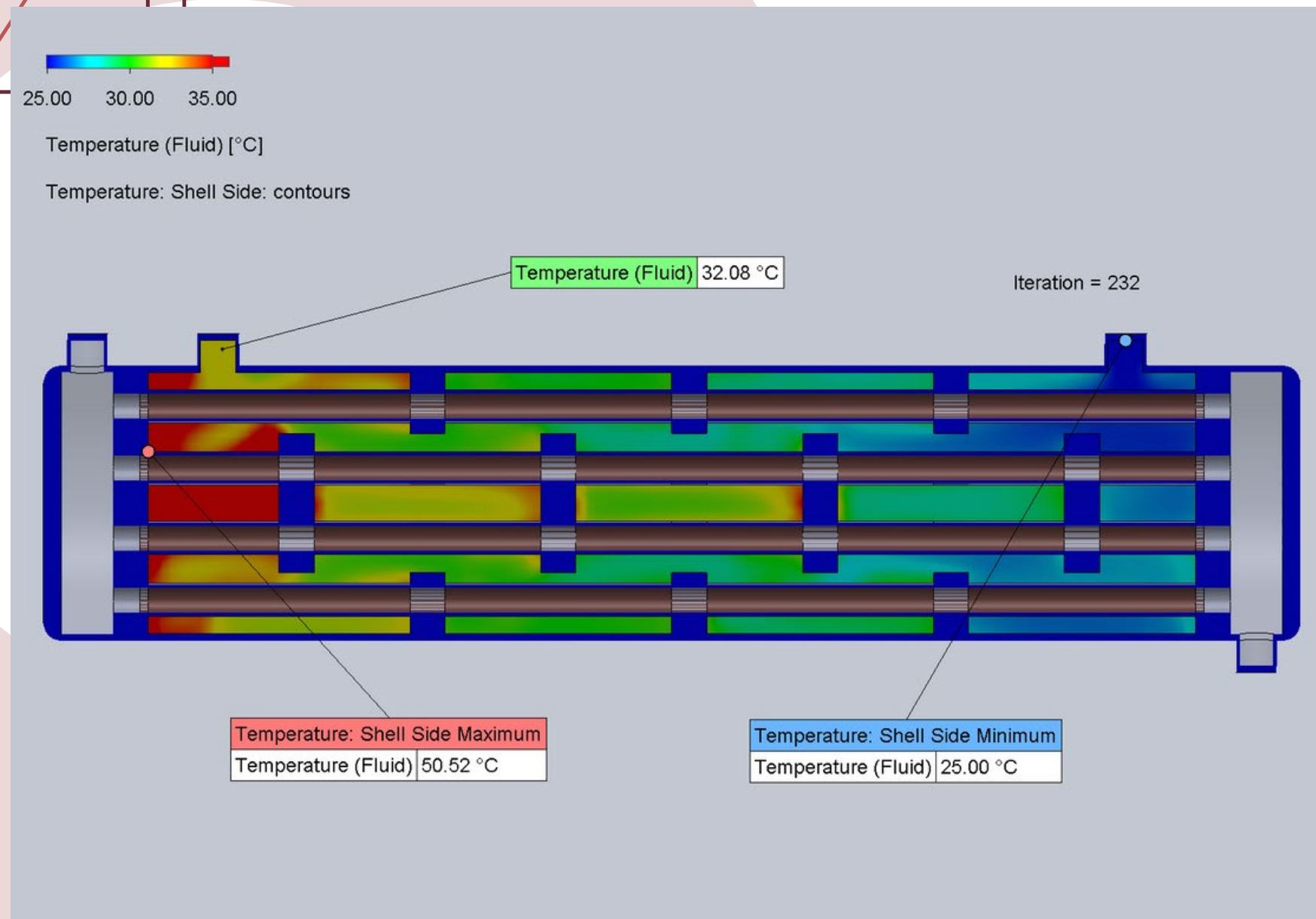
Temperature
Distribution in tube
side



Temperature Output
from Simulation:

Goal plot 1				
Name	Current Value	Progress	Criterion	Averaged Value
Cold inlet Temp	25 °C	Achieved (IT = 95)	2.9815e-06 °C	25 °C
Cold Outlet Temp	32.0268 °C	Achieved (IT = 232)	0.147528 °C	31.9707 °C
Hot Inlet temp	70 °C	Achieved (IT = 95)	3.4315e-06 °C	70 °C
Hot Outlet Temp	62.6729 °C	Achieved (IT = 141)	0.217658 °C	62.672 °C
V max	1.35476 m/s	Achieved (IT = 106)	0.00679709 m/s	1.35583 m/s

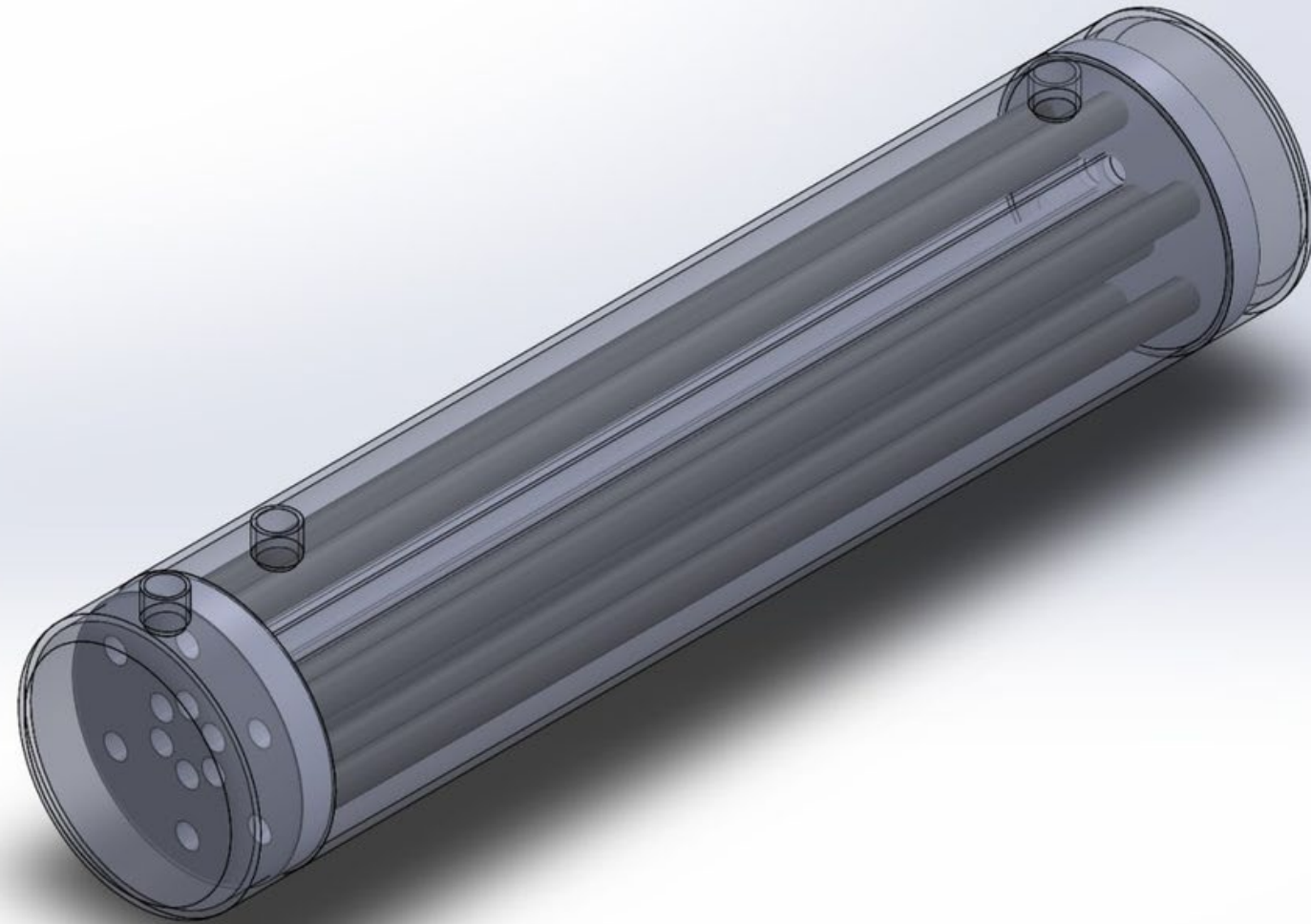
Flow Simulation using SolidWorks



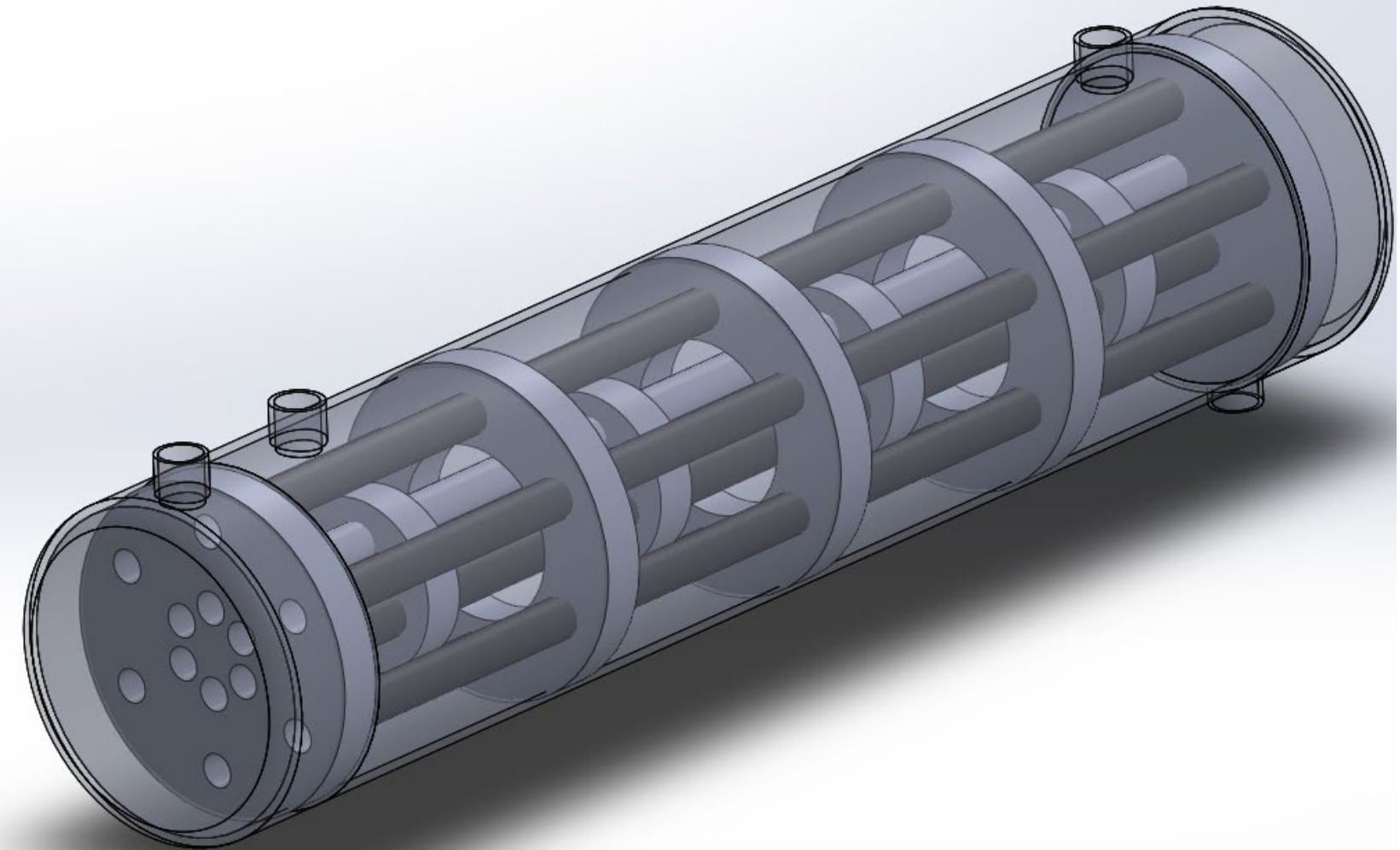
Temperature Output
from Simulation:

Streamline of Flow

Conventional vs. modified baffle configurations



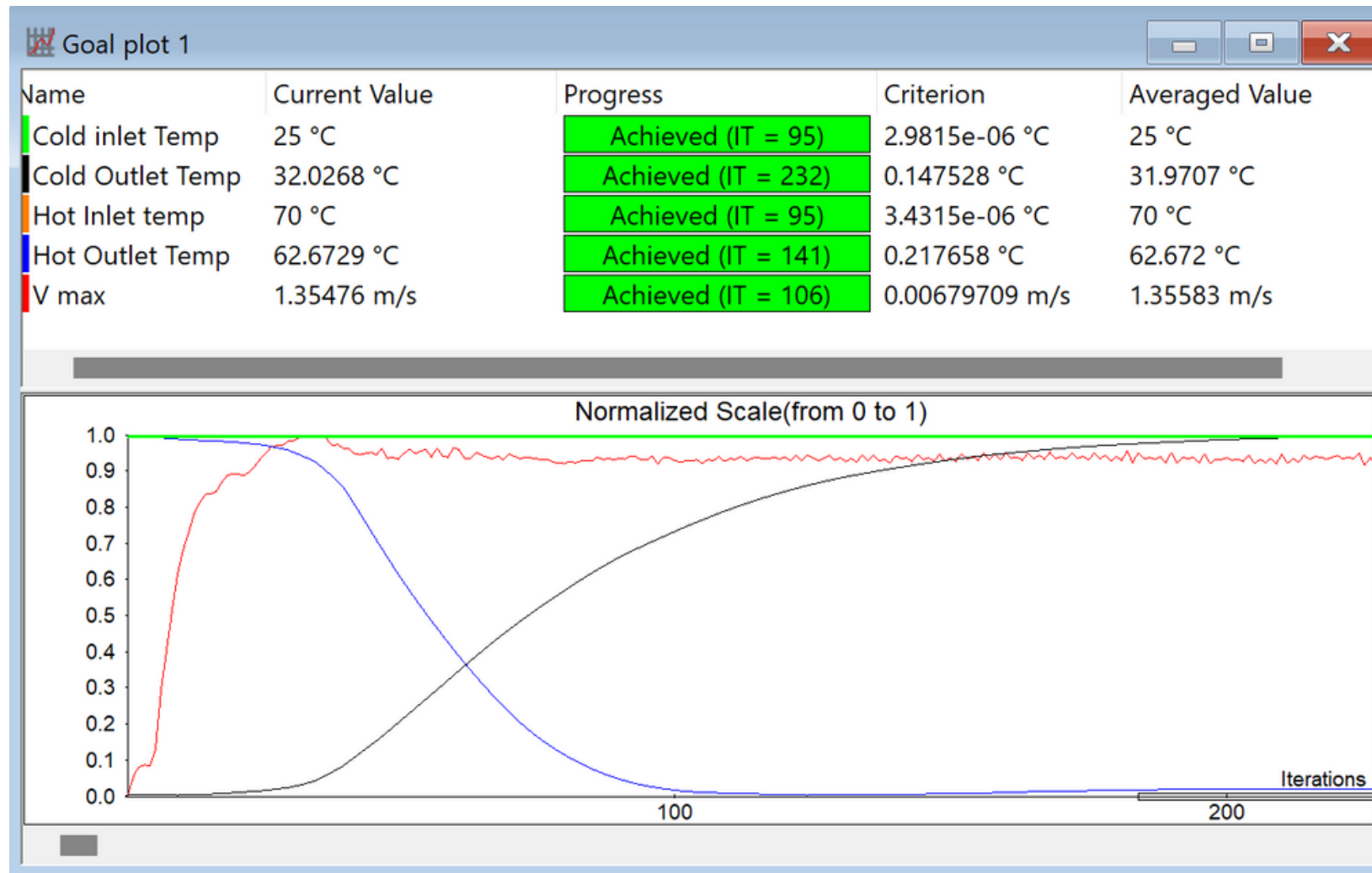
STHX without Baffles



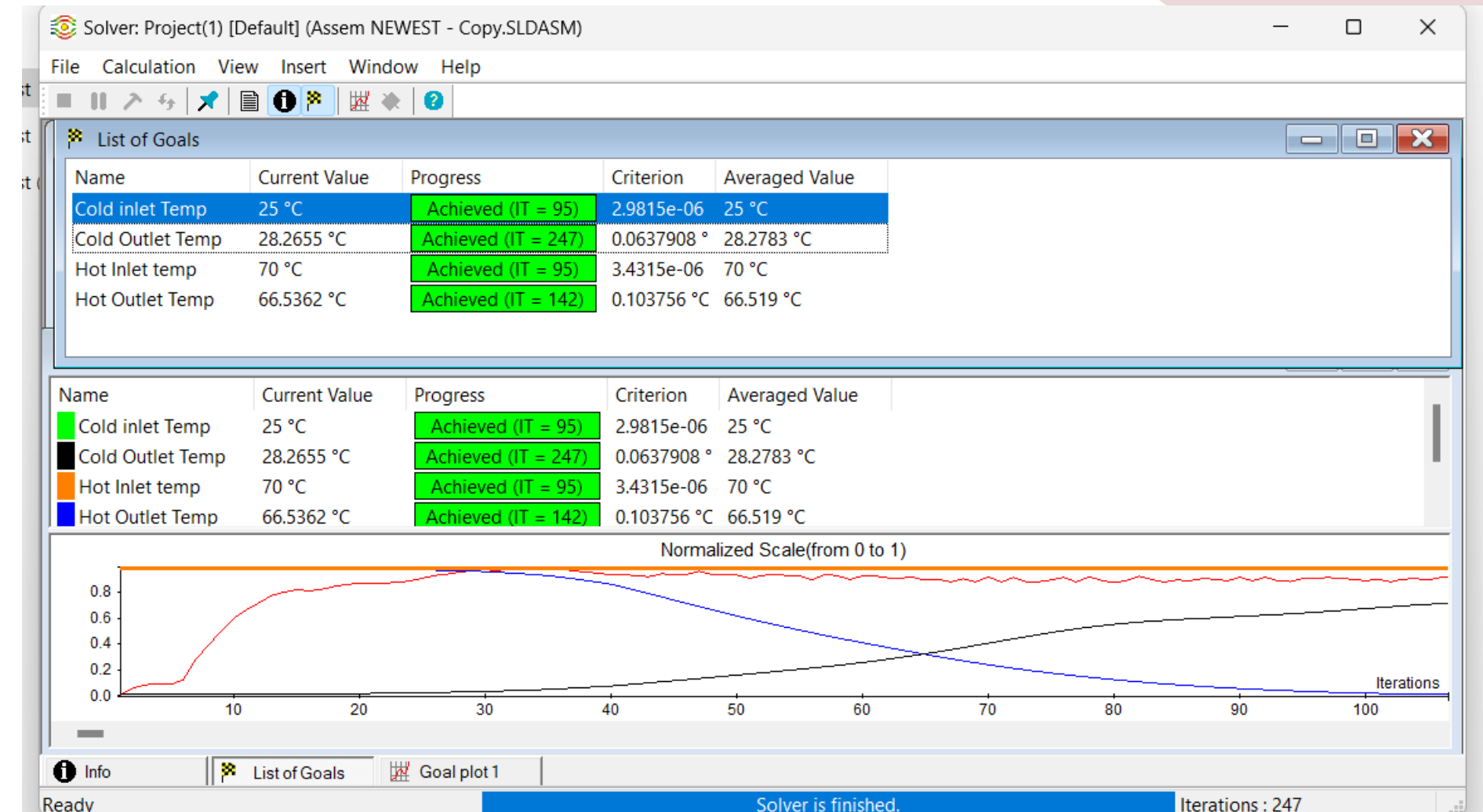
**Modified STHX
with disk and doughnut
baffles**

Performance Comparison

Simulation Results With Disk and Doughnut Shaped Baffles

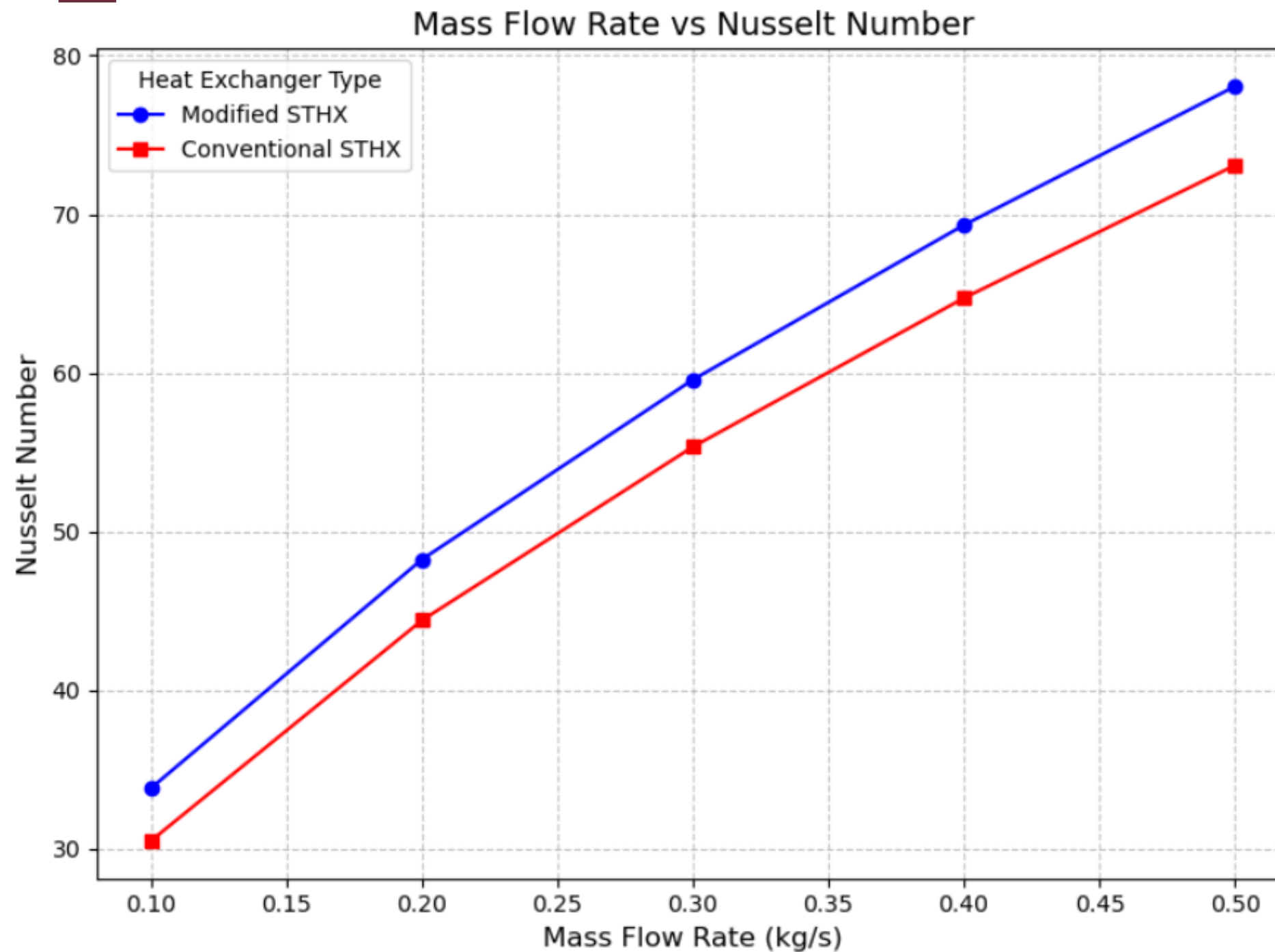


Simulation Results Without Baffles



- Without baffles, the temperature drop at the hot outlet is approximately 66.5°C. However, with our proposed disk-and-doughnut -shaped baffles, the temperature drop at the hot outlet is reduced to around 62°C.
- This indicates that the modified STHX provides improved thermal performance .

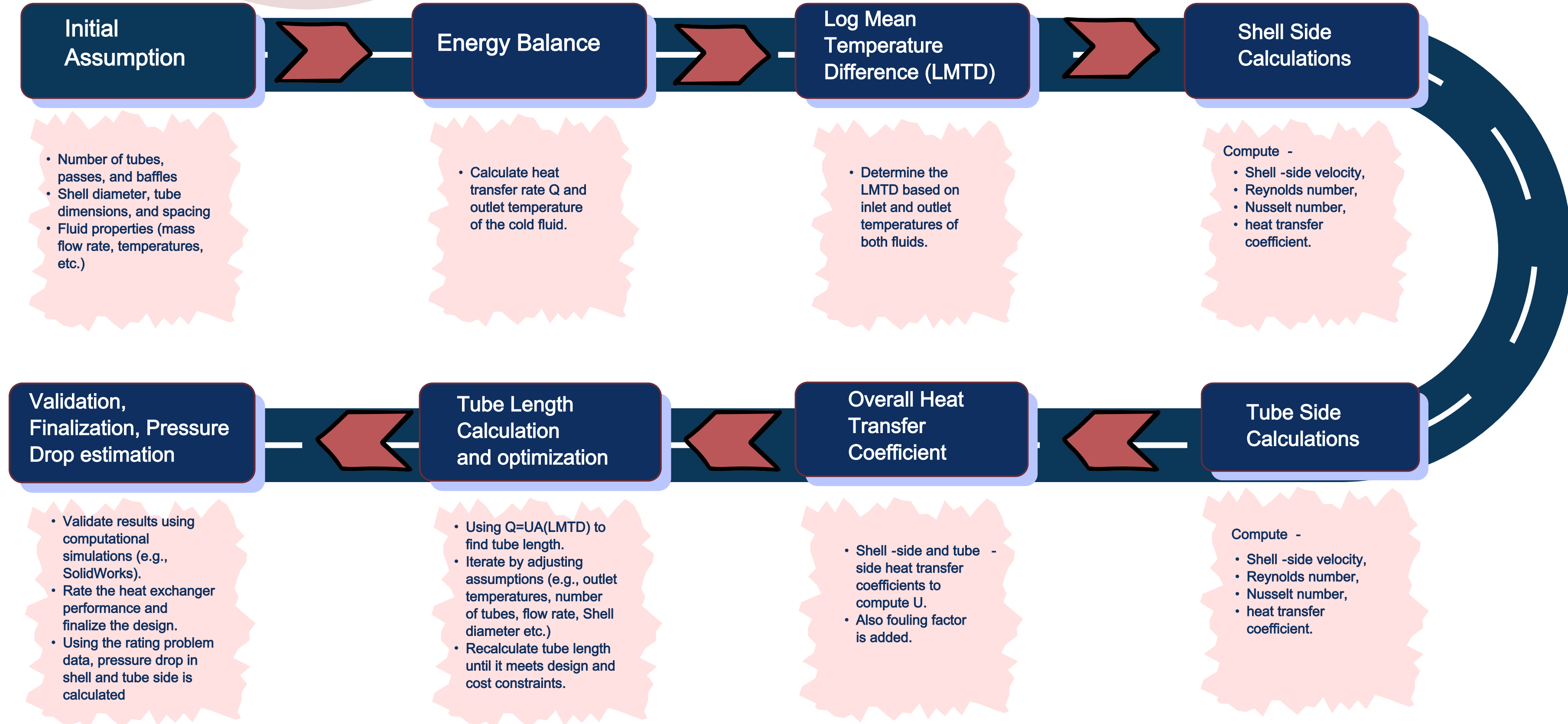
Efficiency Comparison



Nusselt Number Comparisons
(Conventional Vs Modified STHX)

- STHX with **disk and doughnut baffles** consistently produces a **higher Nusselt number** compared to the conventional STHX at any given mass flow rate
- So, Modifications implemented in the heat exchanger design have resulted in a more effective heat transfer process .

Mathematical Calculation Flowchart



Different Stages of Manufacturing



Pressure and Leakage Test

After fabricating the STHX, we tested it for leakage and pressure resistance at the BUET fluid testing lab. Our project successfully passed the test, as shown in the attached picture. No leakage was detected during the pressure test, and it withstood nearly 300 psi without any issues. Additionally, there was no leakage on either the shell side or the tube side.



Group No.05
Section: 82
Student ID:
2010093
2010091
2010119
2010104
Project Name:
Design and fabrication of a Shell and Tube Heat Exchanger
with Disk and Doughnut Shaped Baffles.

Pressure test: 300 psi sustained
no leakage was observed
Time: 17-12-24
Sign: M. L. K.
Received by Fluid Lab.
ME, BUET
Date: 17-12-24
Time: 3:00 PM
Sign: M. L. K.

Pressure Test certified by BUET

Group No. 5
Section: 82
Student ID:
2010093
2010094
2010119
2010104
Project Name:
Design and Fabrication of a Shell and Tube
Heat Exchanger with Disk and Doughnut shaped
Baffles
Received by Fluid Lab.
Date: 15-12-24
Time: 15-12-24
Sign: M. L. K.

Flow Test certified by BUET

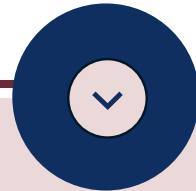
Design Specification

Material	Copper (tube) and Mild Steel (shell)
Tube	3/4 BWG 16
Tube Outer diameter	0.018m
Tube Inner Diameter	0.016m
Number of tube	10
Pitch	1 inch (0.0254m)
Shell diamter	6 inch (0.15m)
Baffle spacing	3inch (0.076m)
Number of Baffle	6
Number of passes	1
Tube Length	24in (0.6m)

Cost Analysis

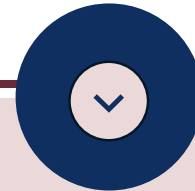
	Unit	Cost per unit	Cost
Shell	1	Tk. 2500	Tk.2500
Tubes	10	Tk. 150	Tk. 1500
Baffles	6	Tk. 400	Tk. 2400
Covers	2	Tk. 500	Tk. 1000
socket	4	Tk. 50	Tk. 200
Base	1	Tk. 400	Tk. 400
Assembly	1	Tk. 3000	Tk. 3000
Total cost			Tk. 11,000

Limitations



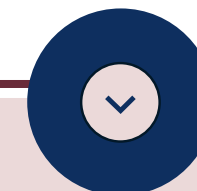
Complex Design

The design and manufacturing process involved increased complexity.



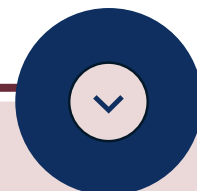
Stress Neglected

Stress analysis for the baffles was not conducted.



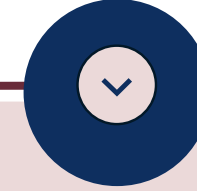
Limited Testing

Flow conditions were not varied to assess performance under different operating scenarios, and testing under real-world conditions was limited.



Operational Uncertainty

Performance under diverse operating scenarios was not fully evaluated.



Repair Constraints

The designed prototype lacked sufficient provisions for repair and maintenance in the inner section due to the fully welded structure.

Acknowledgements

This project was supported by the Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET). We extend our heartfelt gratitude to the following individuals for their invaluable guidance and support :

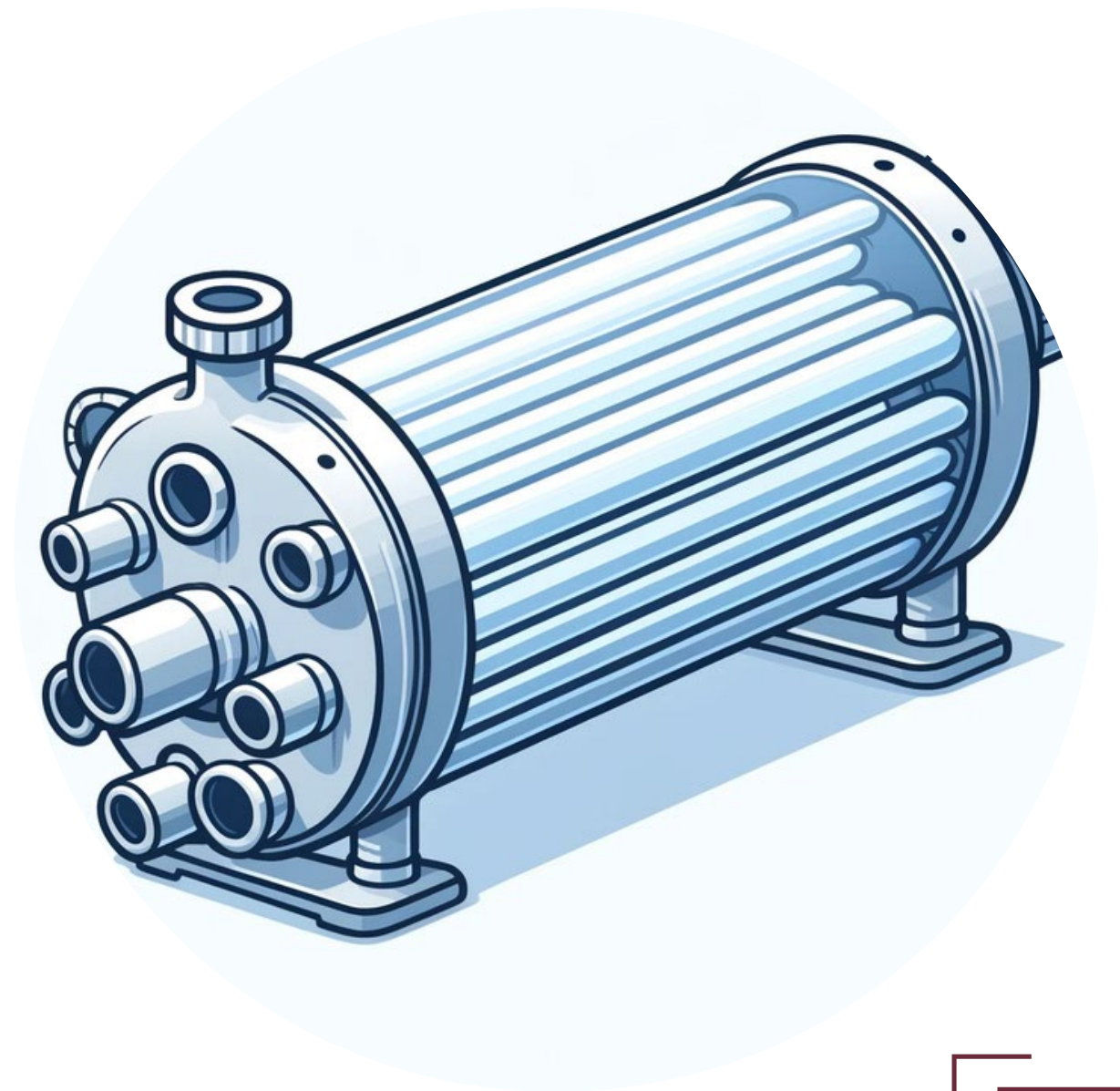
- Mr. Saifuddin Zafar (Lecturer, Dept. of Mechanical Engineering, BUET)
- Mr. Sakib Javed (Lecturer, Dept. of Mechanical Engineering, BUET)
- Mr. Priom Das (Lecturer, Dept. of Mechanical Engineering, BUET)

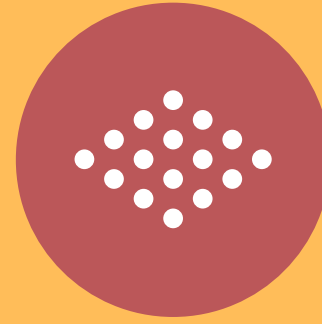
We are also thankful to the BUET Hydraulics Lab and its instructors for their crucial support during the testing phase, which was instrumental in validating our constructed model .

Conclusion



A disk-and-doughnut-shaped baffle was designed for use in shell-and-tube heat exchangers (STHXs). CFD analyses were then performed to evaluate and compare the performance of heat exchangers equipped with conventional baffles and the proposed disk-and-doughnut-shaped baffles. The final manufactured baffle included slight design modifications, and its performance was also analyzed to compare with the previously studied baffle configurations. This comprehensive analysis highlighted the performance differences and potential advantages of the refined baffle design. The manufactured heat exchanger performed excellently in both pressure and flow tests conducted in BUET and no leakage was detected.





**THANK
YOU!**



QUESTION